The Geometry of High-Dimensional Safety: Evaluating Refusal Cones in Extended Refusal Language Models

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Abstract

Refusal behavior in aligned language models can often be traced to a single direction in the residual stream, which abliteration attacks exploit via weight projection. Abu Shairah et al. (2025) propose Extended Refusal (ER), a fine-tuning defense that trains models to produce structured explain–refuse–justify responses, and argue via PCA centroid analysis that this distributes refusal across the activation space. We reproduce ER on Qwen2.5-3B-Instruct using LoRA and evaluate it against abliteration and TwinBreak. We then audit the resulting representations using the gradient-based framework of Wollschläger et al., measuring DIM–RDO alignment, refusal cone dimensionality, and per-basis causal load. ER preserves robustness against standard abliteration with minimal utility cost, and the maximum viable refusal cone expands from dimension 2 in the baseline to dimension 5 in three of four ER variants. However, within-cone analysis shows that ER-full's high-dimensional cone is functionally near-rank-1: a single basis direction carries most of the ablation effect. A gradient-based cone attack breaks ER-full more effectively than standard abliteration. The form of safety expression in training data measurably reshapes refusal geometry, but current ER training does not produce a defense robust to a geometrically-aware attacker.

1 Introduction

As large language models (LLMs) have grown more capable, ensuring safety through alignment has become increasingly important. Alignment methods train models to refuse harmful prompts but prior work has shown that the refusal signal can be isolated and suppressed through simple linear transformations of model weights without any retraining. The success of these lightweight attacks suggests that current safety alignment is not robust to weight projections.

To address this, Abu Shairah et al. (2025) propose Extended Refusal (ER), a fine-tuning method that restructures refusals into a three-part format. The intuition is that this extended format distributes the refusal signal more broadly across model weights, making it harder to isolate via linear interventions.

In this study, we reproduce the findings of Abu Shairah et al. (2025) by fine-tuning Qwen2.5-3B-Instruct with ER and evaluating its robustness against abliteration. We then extend their work by deploying an out-of-domain attack, TwinBreak (Krauß et al., 2025), and by scrutinizing the geometric mechanics behind the defense in greater depth. For this analysis, we draw on Refusal Direction Orthogonality (RDO) and the Refusal Cone (Wollschläger et al., 2025) to more precisely characterize how refusal representations are structured, allowing us to test whether ER genuinely disperses the safety signal across a high-dimensional subspace.

We hypothesize that the brittleness of aligned refusal is not a fundamental property of refusal behavior but a consequence of alignment procedures that reinforce short, formulaic refusal patterns. If refusal is concentrated in a single dominant direction, projection-based attacks should remain effective. Conversely, if ER distributes refusal such that no single direction carries the bulk of the causal load, geometric attacks should fail across the entire refusal cone — not only against the specific direction DIM identifies. The geometric framework distinguishes two interpretations of any observed robustness: a genuinely distributed defense versus a relocated single direction.

We reproduce ER's behavioral claim under LoRA fine-tuning: ER substantially improves robustness to abliteration while preserving general utility. Our geometric analysis complicates the mechanistic interpretation. ER training does change refusal geometry — the DIM candidate

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landscape collapses, gradient-optimized directions become misaligned with mean-difference directions, and the refusal cone expands from dimension 2 in the baseline to dimension 5 in ER-full. But the within-cone causal load is concentrated: one basis direction carries most of the ablation effect, and a cone-based attack breaks ER-full more effectively than standard abliteration. The defense is real, but it is better described as making single-direction extraction harder than as distributing refusal across independent mechanisms.

2 Background and Motivation

Modern LLMs are typically aligned through a multi-stage post-training pipeline designed to suppress harmful behaviors, primarily through supervised fine-tuning (SFT) and reinforcement learning from human feedback (RLHF). In SFT, the model is trained on pre-labeled instruction-response pairs that include harmful requests paired with refusal responses and is optimized using next-token prediction loss to imitate these examples. RLHF further refines this behavior by training a reward model on ranked human preferences and optimizing the language model accordingly (Christiano et al., 2023). However, because both methods repeatedly reinforce similar refusal patterns (e.g., "I cannot help you with that"), refusal behavior can be compressed into a narrow representational pathway within the residual stream rather than being distributed across the network. Recent work (Arditi et al., 2024) suggests that this creates a geometric brittleness because refusal behavior that depends disproportionately on a low-dimensional subspace can have its safety alignment easily suppressed.

2.1 Abliteration: Suppressing Refusal via Weight Orthogonalization

Arditi et. al. demonstrated that refusal behavior in LLMs is largely mediated by a single linear direction in the residual stream, across 13 open-source models up to 72B parameters. They rely on the difference-in-means calculation, in which the mean residual stream activations are calculated for n harmful prompts $\mathcal{D}^+$ and m harmless prompts $\mathcal{D}^-$ for each layer ℓ and post-instruction token position i:

$$\mu_i^{(\ell)} = \frac{1}{n}\sum_{t\in\mathcal{D}^+} x_i^{(\ell)}(t), \qquad \nu_i^{(\ell)} = \frac{1}{m}\sum_{t\in\mathcal{D}^-} x_i^{(\ell)}(t), \tag{1}$$

From this, we calculate the candidate direction $r_i^{(\ell)} = \mu_i^{(\ell)} - \nu_i^{(\ell)} \in \mathbb{R}^d$ per (layer, position) pair. The best single direction is then selected as:

$$\hat{r} = \underset{r_i^{(\ell)}}{\arg\max}\ \Delta_{\text{refusal}}(M, r_i^{(\ell)}), \tag{2}$$

where Δ_{refusal} quantifies how ablating the direction drops refusal on harmful prompts, and how much adding it induces refusal on harmless prompts. $\hat{r}$ is then normalized.

2.1.1 Causal Verification

Arditi et al. show that $\hat{r}$ is necessary for refusal. At the time of inference, directional ablation zeroes out the $\hat{r}$ from every residual stream activation at every layer and position:

$$x' \leftarrow x - \hat{r}\hat{r}^\top x. \tag{3}$$

This prevents the model from ever representing $\hat{r}$, collapsing refusal rates. Conversely, adding $r^{(\ell^*)}$ at the selected layer induces refusal on harmless prompts.

2.1.2 Weight Orthogonalization

Rather than applying Eq. (3) at every forward pass, the intervention can be included permanently in the weights. For each matrix $W_{\text{out}} \in \mathbb{R}^{d\times d_{\text{in}}}$ that writes to the residual stream (attention output projections, MLP output projections, embedding matrix) the $\hat{r}$ component can be projected out:

$$W'_{\text{out}} \leftarrow \underbrace{\left(I_d - \hat{r}\hat{r}^\top\right)}_{P_{\hat{r}}} W_{\text{out}}, \tag{4}$$

where $P_{\hat{r}}$ is the orthogonal projector onto $\hat{r}^\perp$. This is exactly equivalent to inference-time ablation, where for any input x, we can find $W'_{\text{out}}x = (I - \hat{r}\hat{r}^\top)(W_{\text{out}}x)$. Applying Eq. (4) to every matrix in the residual stream yields an abliterated model $\tilde{M}$ that cannot represent $\hat{r}$ in its residual stream.

2.2 Extended Refusal

Abu Shairah et al. (2025) proposed Extended Refusal (ER) as a defense against abliteration. Rather than training models to produce short refusals, ER fine-tunes models to generate structured three-part responses to harmful prompts: an **explanation** of why the request is unsafe, an explicit **refusal**, and an ethical **justification**. The core intuition is that this response format distributes refusal behavior across a broader representational subspace, making it harder to suppress through low-rank interventions such as abliteration.

2.3 Limitations of PCA-Based Geometric Analysis

To support the hypothesis that Extended Refusal (ER) distributes safety representations, Abu Shairah et al. (2025) analyze residual stream activations using Principal Component Analysis (PCA). In baseline models, harmful and harmless prompts separate strongly along a dominant principal component, whereas ER fine-tuning produces more diffuse separation. The authors quantify this effect using centroid distance in PCA space:

$$\delta = \|\bar{h}_{\text{harmful}} - \bar{h}_{\text{harmless}}\|_2 \tag{5}$$

They find that centroid distance collapses after abliteration in baseline models but remains comparatively preserved under ER, which is interpreted as evidence of more distributed refusal representations.

However, PCA is descriptive rather than causal. Principal components capture directions of maximal variance, not necessarily the directions that mediate refusal behavior. A model may preserve centroid distance while still depending on a small number of intervention-sensitive refusal directions. From an attacker's perspective, the relevant question is not whether harmful and harmless activations remain separated, but whether any low-rank intervention can suppress refusal.

We therefore require a framework that directly measures how refusal behavior is distributed across causally effective directions in representation space.

3 Geometric Framework (Analytical Setup)

The PCA-based centroid distance analysis used by (Abu Shairah et al., 2025) describes how harmful and harmless prompt activations are distributed, but does not identify which directions in the residual stream causally mediate refusal behavior. A direction may explain little variance while still carrying most of the refusal signal. Since our central question is whether Extended Refusal (ER) redistributes safety across multiple independent mechanisms, we require a causal geometric framework rather than a descriptive one.

We adopt and extend the framework of (Wollschläger et al., 2025), combining Difference-in-Means (DIM), Refusal Direction Optimization (RDO), Representational Independence (RepInd), and refusal cone analysis to characterize how ER fine-tuning reshapes refusal geometry.

3.1 Difference in Means (DIM)

We begin with the Difference-in-Means (DIM) method introduced by (Arditi et al., 2024). For each layer ℓ and token position i, we compute a candidate refusal direction from the difference between mean harmful and harmless residual activations:

$$r_i^{(\ell)} = \mu_i^{(\ell)} - \nu_i^{(\ell)}, \tag{6}$$

where $\mu_i^{(\ell)}$ and $\nu_i^{(\ell)}$ are defined in Eq. 1. Candidate directions are evaluated by whether ablating them suppresses refusal on harmful prompts, adding them induces refusal on harmless prompts, and ablation preserves benign behavior (low KL divergence).

DIM provides an important baseline because it assumes refusal is concentrated in a single dominant direction. If ER redistributes refusal across multiple mechanisms, DIM should recover weaker or less stable intervention directions.

3.2 Refusal Direction Optimization (RDO)

Because DIM may fail when refusal is distributed, we apply Refusal Direction Optimization (RDO) (Wollschläger et al., 2025), which directly optimizes a refusal direction for causal effectiveness.

RDO refines a candidate direction by jointly optimizing three objectives:

$$\mathcal{L}_{\text{RDO}} = \lambda_{\text{abl}}\mathcal{L}_{\text{ablation}} + \lambda_{\text{add}}\mathcal{L}_{\text{addition}} + \lambda_{\text{ret}}\mathcal{L}_{\text{retain}}, \tag{7}$$

where $\mathcal{L}_{\text{ablation}}$ encourages harmful compliance under ablation, $\mathcal{L}_{\text{addition}}$ induces refusal on harmless prompts, and $\mathcal{L}_{\text{retain}}$ penalizes degradation of benign behavior through KL divergence.

Unlike DIM, RDO explicitly trades off intervention strength against side effects, making it especially relevant for ER variants where refusal may no longer align with the dominant harmful–harmless contrast. We compare DIM and RDO directions across ER variants to test whether ER alters refusal geometry sufficiently to invalidate simple difference-in-means extraction.

3.3 Representational Independence

A single refusal direction is insufficient if refusal behavior is distributed. We therefore test whether additional directions independently mediate refusal or merely appear functional due to entanglement with previously discovered directions.

Following (Wollschläger et al., 2025), we use representational independence (RepInd) which is stronger than orthogonality. Two directions are RepInd if ablating one does not alter the cosine alignment of the other across residual activations. We iteratively train RepInd directions, requiring each new direction to remain independent of all previously identified refusal directions. The number of above-chance RepInd directions serves as a lower-bound estimate of how many independent refusal mechanisms a model contains.

This quantity directly tests our hypothesis: if ER improves robustness through representational dispersion, ER variants should exhibit more RepInd directions than the baseline.

3.4 Cone Dimensionality

If multiple independent refusal directions exist, they span a higher-dimensional refusal subspace. Following (Wollschläger et al., 2025), we model this structure as a refusal cone:

$$\mathcal{C}_N = \left\{ \sum_{j=1}^{N} \alpha_j b_j \;\middle|\; \alpha_j \geq 0 \right\} \setminus \{0\}, \tag{8}$$

where $B = [b_1, \ldots, b_N]$ is an orthonormal basis of refusal directions.

We optimize cone bases using Refusal Cone Optimization (RCO), jointly training basis vectors under the RDO objective while enforcing orthogonality. Cone quality is evaluated by sampling directions from the cone and measuring ablation ASR.

Cone dimensionality provides a complementary estimate of refusal complexity: RepInd gives a conservative lower bound through sequential discovery, while cone optimization provides an upper-bound estimate constrained by optimization difficulty. Comparing these quantities across baseline and ER variants allows us to test whether ER increases the effective dimensionality of refusal behavior.

4 Methodology

To rigorously evaluate how the ER defense distributes safety across independent latent mechanisms or only manipulates a single refusal direction, we designed a multi-stage pipeline that combines reproduction of the original ER claims with two extensions: advanced behavioral attack (TwinBreak) and geometric analysis as detailed in section 3.

4.1 Experimental Setup

Dataset. We use the ER dataset (train split) provided by Abu Shairah et al. (2025), which contains 4,268 harmful ER examples and 5,732 benign Alpaca-GPT4-en examples (Wang et al., 2022) for a total of 10k samples.

Models. Following Abu Shairah et al. (2025), we fine-tune Qwen2.5-3B-Instruct (Team, 2024) as our base model using supervised fine-tuning (SFT). To isolate the contribution of each component in the ER format, we create four ablations: the baseline model fine-tuned on the refusal segment of the dataset examples only, justification only, explanation only and the full ER dataset.

We split each harmful response using key marker phrases to create these ablations. We start the refusal segment at the first occurrence of "I'm sorry," and the justification segment at the phrases "ethical AI", "ethical considerations" or "ethical principles." Text preceding the refusal marker is treated as the explanation segment. We found that this heuristic recovered all three segments for 4,168 of 4,268 harmful examples (97.8%). In 50 cases (1.2%), the ethical rationale did not match the justification markers and was kept with the refusal segment. Described more thoroughly in our limitations section, we recognized that this approach may introduce a limited bias favoring the refusal-only condition.

We therefore consider five model variants in total: (1) SSFT baseline, (2) ER-full, (3) ER-explanation-only, (4) ER-justification-only and (5) ER-refusal-only.

Fine-tuning. Given computational constraints, we use parameter-efficient Low-Rank Adaptation (LoRA) (Hu et al., 2022) with rank = 64 and alpha = 128 rather than the full fine-tuning employed by Abu Shairah et al. (2025). We acknowledge that differences in ER defense effectiveness may arise from this lower-rank parameterization and discuss these in the limitations section. Fine-tuning was performed on a single A5000 GPU with 8 CPUs and 64GB RAM, taking approximately 1.5 hours per ablation over 3 epochs with a learning rate of 1.0×10^{-6}, replicating the approach from Abu Shairah et al. (2025).

4.2 Evaluation Metrics

With our work, we bridge two lines of research with different evaluation protocols. Wollschläger et al. (2025) measure Attack Success Rate (ASR) using the StrongREJECT fine-tuned judge (Souly et al., 2024) on JailbreakBench (Chao et al., 2024), while (Abu Shairah et al., 2025) measure refusal rate using Qwen-2.5-14B as an LLM-as-judge on CatQA (Bhatt et al., 2024).

We adopted a two-track evaluation framework that keeps each paper's original approach and provides a shared surface for comparison. Geometry experiments are scored with StrongREJECT on JailbreakBench and defense experiments are scored with Qwen-2.5-14B on CatQA. All extended-refusal ablation variants are additionally evaluated on JailbreakBench with StrongREJECT, so that directions discovered on baseline and ER fine-tuned models can be compared under the same prompts and scoring. More details on the safety and utility metrics can be found in the Appendix.

4.3 Reproduction of Baseline Abliteration

We first reproduced the experimental setup of (Abu Shairah et al., 2025) to establish a control baseline to test whether ER neutralizes standard abliteration by distributing the refusal signal. We applied standard weight-orthogonalization (linear abliteration)(Arditi et al., 2024) to each of the five variants. This attack assumes refusal is mediated by a single, easily identifiable principal component in the residual stream and surgically suppresses it to unalign the model. Each abliterated variant was produced on 1 NVIDIA A100 GPU in 1 hour. The full pipeline including pre- and post-attack evaluation completed in approx. 4.5 hours per variant.

4.4 Extension via Advanced Threat Model

While standard abliteration tests linear vulnerability, it cannot determine whether a defense is structurally robust or merely shifting the safety vector outside the PCA distribution. To close this gap, we subjected all variants to TwinBreak (Krauß et al., 2025), a modern non-directional attack that uses high-similarity "twin prompts" to isolate and prune the specific network parameters responsible for safety filtering. Following the original TwinBreak schedule (Krauß et al., 2025), we pruned 1% of ranked hidden dimensions per round (after utility exclusion) over 5 cumulative rounds, as the authors assert this balances progressive discovery of distributed safety parameters. We retained the top 0.1% of dimensions ranked by harmless-vs-harmless twin gaps as utility parameters. Each attack ran on 1 NVIDIA A100 GPU in under two minutes and the full evaluation pipeline took approx. 5 hours per variant.

4.5 Extension via Geometric Probing

To map the true dimensionality of the refusal cone enabled by ER finetuning, we applied Refusal Direction Optimization (RDO), drawn from Wollschläger et al.'s (Wollschläger et al., 2025) mechanistic framework, to all five model variants. For each model we: (a) extracted DIM directions across all (layer, position) pairs and selected the optimum using the Arditi et al. (Arditi et al., 2024) heuristic; (b) trained an RDO direction at the same layer as the selected DIM direction with hyperparameters from Wollschläger et al. (Wollschläger et al., 2025) ($\lambda_{\text{abl}} = 1.0$, $\lambda_{\text{add}} = 0.2$, $\lambda_{\text{ret}} = 1.0$; AdamW, $\text{lr} = 0.01$ with plateau-based decay; 256 effective samples per batch); (c) iteratively trained up to five RepInd directions, each constrained to be independent of all previously discovered directions and of the DIM direction ($\lambda_{\text{ind}} = 200$; RepInd loss computed over the first 90% of layers); and (d) trained concept cones of dimensionality 2–6 via Refusal Cone Optimization with Gram–Schmidt re-orthogonalization between gradient steps. For each cone we sampled 256 unit vectors uniformly from the cone interior and report both median and lower-quartile ablation ASR on JailbreakBench (Chao et al., 2024). All geometry experiments ran on 1 NVIDIA A100 GPU and took an estimated of 7 hours per model ablation.

5 Behavioral Results

5.1 Standard Abliteration

Abliteration collapses the base model refusal rate from 0.866 to 0.304 which aligns with findings that standard SFT concentrates refusal behavior into removable directions (Abu Shairah et al., 2025). Despite a lower pre-attack refusal rate than three ablation variants, ER-Full maintains a post-attack rate of 0.873, 56.9 percentage points higher than that of the baseline model.

Surprisingly, the refusal-only variant proved most robust to abliteration, with refusal dropping only from 0.96 to 0.90 and achieving the highest refusal rate indicating that its refusal behavior resists isolation. This contrasts with prior

Fine-tuning Condition	Pre-Attack			Abliteration			TwinBreak		
	Refusal↑	MMLU↑	PPL↓	Refusal↑	MMLU↑	PPL↓	Refusal↑	MMLU↑	PPL↓
Baseline	0.866	0.663	**13.83**	0.304	0.643	15.85	0.004	**0.473**	**21.38**
ER-justification only	**0.971**	0.673	14.83	0.802	**0.665**	17.63	0.567	0.448	34.79
ER-explanation only	0.705	0.674	14.88	0.724	0.652	16.29	**0.769**	0.426	33.86
ER-refusal only	0.960	**0.675**	14.90	**0.900**	0.648	**15.49**	0.689	0.403	37.18
ER-full	0.842	0.673	14.87	0.873	0.652	17.26	0.720	0.403	37.18

Table 1: Behavioral robustness and utility across model variants before attack, after abliteration, and after TwinBreak. We report refusal rate, MMLU accuracy, and C4 perplexity.

work (Abu Shairah et al., 2025), suggesting a representational difference between LoRA and full fine-tuning: full fine-tuning may rewrite weights sufficiently to form a dominant refusal direction, whereas LoRA redirects refusal behavior already distributed across frozen base weights.

Explanation-only training produced minimal refusal change (0.705 to 0.724), though its rate still falls below ER-Full, supporting the necessity of additional refusal components. Justification-only training exhibited the largest degradation among ER ablations, suggesting that ethical reasoning alone can yield a relatively concentrated and therefore vulnerable refusal mechanism.

Most notably, these robustness gains come at negligible utility costs: MMLU shifts by at most 2.7 percentage points (ER-refusal-only) and C4 perplexity increases by no more than 2.8 points (ER-justification-only) across all conditions, confirming that abliteration selectively suppresses safety behavior while preserving general capabilities.

5.2 Stress-Testing with TwinBreak

TwinBreak exposes a qualitatively different vulnerability than abliteration. The baseline refusal rate is reduced to near-zero (0.004) under TwinBreak, significantly below abliteration's 0.304, confirming that parameter-level pruning bypasses safety mechanisms that directional removal cannot fully suppress. ER-full maintains a refusal rate of 0.720, a 14.5-point drop from pre-attack versus only 3.1 increase under abliteration. While ER's directional dispersal is only partially effective against parameter-level attack (Twinbreak), we show that it still provides meaningful residual robustness even outside its intended threat model.

Component rankings also shift between attacks: ER-refusal-only leads under abliteration (0.900), while ER-explanation-only leads under TwinBreak (0.769). Since TwinBreak prunes MLP parameters with divergent activations across matched harmful–harmless pairs, rather than removing a single residual-stream direction, the explanation component's safety pathways are likely harder to isolate through activation-difference pruning. ER-full never tops either attack alone but offers the most balanced defense across both, which helps to support a complementary relationship of the three-part format.

Utility costs under TwinBreak are significant across all ablations: MMLU drops to $\sim$0.40 and perplexity rises to 33-37 for ER variants, versus 0.473 and 21.38 for the baseline. ER models pay a steeper price because their safety signal is more entangled with core computation paths. Therefore, an attacker who strips safety from an ER-defended model produces a noticeably less capable model, which can serve as a potential signal of compromise.

6 Mechanistic Analysis

Section 5 establishes that ER helps to preserve refusal under attacks. We now ask whether this robustness arises from the geometric structure of refusal representations, as Abu Shairah et al. (2025) hypothesize. We measure four quantities for each model variant: the DIM candidate landscape, the cosine alignment between DIM and gradient-optimized directions, the maximum cone dimensionality supporting above-chance ablation, and the per-basis causal load within recovered cones.

The geometric hypothesis predicts that ER variants exhibit (i) higher cone dimensionality than the baseline, (ii) rankings by post-attack refusal correlated with rankings by cone dimensionality and (iii) per-basis ablation ASR bounded away from concentration in a single dominant basis. The alternative is that ER's robustness reflects difficulty of direction identification rather than geometric distribution. This would manifest as low DIM-RDO alignment without corresponding cone expansion, or as nominally high-dimensional cones with one dominant basis.

6.1 DIM landscape under ER fine-tuning

Before measuring cone dimensionality, we examine whether DIM already shows signs of disruption under ER training. Figure 1 visualizes the three DIM selection criteria across all (ℓ, p) candidate pairs for each variant.

The baseline shows a sharply structured viable region. Ablation effects concentrate in mid-layers (18–25), with the selected direction at refusal score -9.55 (Table 2). ER-full, ER-explanation and ER-justification show substantial collapse as ablation columns are uniformly pale, and selected directions are $2\times$ to $5\times$ weaker. ER-explanation's selection at the final layer (35) suggests no usable mid-layer candidate survives the criteria. ER-refusal-only preserves the baseline heatmap structure almost completely, though its selected direction's refusal score (-2.882) is still meaningfully weaker than the baseline's (-9.545).

Training on refusal-only data leaves the DIM landscape pattern largely intact, while explanation and justification alone collapse it.

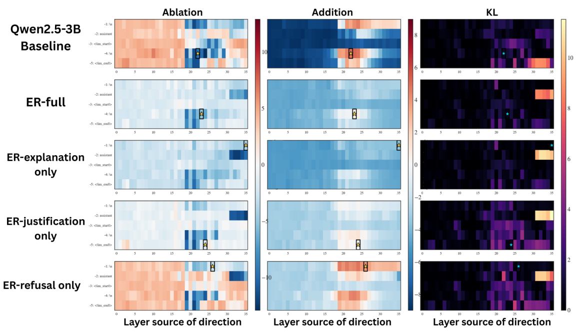

Figure 1: DIM candidate quality across (ℓ, p) pairs for each model variant. Rows correspond to models and columns show ablation score, addition score, and KL divergence under ablation. Stars mark selected directions.

This provides the first evidence for our central mechanistic claim: ER-full disrupts single-direction extraction but this effect is driven primarily by the explanation and justification components. Training on refusal-only data leaves geometry largely unchanged, while explanation and justification alone collapse the DIM landscape. This predicts a corresponding pattern in cone analysis: $N_{\text{ER-refusal}} \approx N_{\text{baseline}}$, while $N_{\text{ER-explanation}}, N_{\text{ER-justification}} > N_{\text{baseline}}$.

We next summarize the selected DIM directions quantitatively in Table 2.

Model	L	P	Tok	Ref	Steer	KL	BL	Abl
Baseline	22	-4	\n	-9.545	4.922	0.808	0.11	1.00
ER-full	23	-4	\n	-5.048	0.535	0.513	0.31	1.00
ER-ref-only	26	-1	\n	-2.882	3.830	0.933	0.00	0.95
ER-exp-only	35	-1	\n	-3.881	-3.647	0.167	0.74	0.74
ER-just-only	24	-5	<|im_end|>	-1.768	0.047	0.682	0.82	0.75

Table 2: DIM-selected directions across model variants.

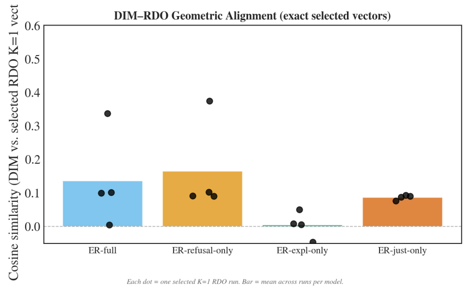

Figure 2: Cosine similarity between DIM-selected and RDO-optimized refusal directions across model variants. All ER variants exhibit near-zero alignment, indicating that RDO identifies a substantially different refusal direction than DIM after ER fine-tuning.

6.2 DIM–RDO Cosine Alignment

If ER moves the refusal direction into a region DIM cannot reach, gradient-based optimization should recover a direction that differs systematically from DIM's selection. Figure 2 reports cosine similarity between DIM-selected and RDO-trained K=1 directions across four independent restarts per variant.

Cosine alignment clusters near zero across all four ER variants (means 0.005–0.166; max 0.38). RDO consistently recovers directions geometrically distinct from DIM's selection, including in ER-refusal-only, where the DIM landscape itself resembled the baseline. ER fine-tuning alters the structure of refusal representations beyond what DIM analysis reveals.

6.3 Cone Dimensionality

We trained refusal cones of dimension $K = 2$ through 5 for each variant and evaluated each by sampling 256 unit vectors and computing median ablation ASR. Figure 3 shows the resulting distributions.

The baseline supports a cone of maximum dimension $K_{\max} = 2$, beyond which the optimizer fails to recover any cone with median ASR above the chance threshold of 0.1. ER-refusal-only similarly has $K_{\max} = 2$, with weaker median ASR (≈ 0.19). ER-full, ER-explanation-only, and ER-justification-only all support $K_{\max} = 5$, with median ablation ASR substantially above chance throughout.

ER training expands the refusal cone for three of four variants — a 2.5× increase over the baseline. ER-refusal-only does not expand the cone, which is consistent with its baseline-like DIM landscape. This confirms prediction (i) directionally, but cone dimensionality alone does not establish causal distribution: a $K = 5$ cone can be effectively rank-1 if one basis carries the bulk of the effect.

6.4 Per-basis causal load

We ablate each basis vector individually within each trained cone and measure StrongREJECT ASR on JailbreakBench. Figure 7 shows per-basis ASR for the best run at each K.

ER-full's cone is causally concentrated. At every K from 1 to 4, basis 0 carries ASR 0.61–0.72 while remaining bases sit below 0.20. The nominally 5-dimensional cone is functionally near-rank-1: cone training succeeds by finding the dominant direction reliably across K, not by distributing across multiple directions.

ER-refusal-only shows no recoverable cone structure: per-basis ASR remains below 0.08 across all bases and all K. ER-explanation-only shows partial distribution at $K = 4$, basis 0 reaches 0.50 with secondary bases at 0.11-0.16. ER-justification-only's cone is non-causal under StrongREJECT (max basis ASR 0.04 at any K), despite substring matching suggesting ASR 0.58–0.78 — a 15–20× divergence that we attribute to the model's hedged, policy-style refusals defeating keyword-based judges.

Only ER-explanation-only shows distributed causal load. ER-full's geometric expansion does not translate into causal distribution. Figure 7 (Appendix) shows per-basis ablation ASR for the best run at each K.

6.5 Vulnerability to RDO-Guided Attacks

Figure 4 compares four attack conditions: no attack, DIM ablation, best K=1 RDO ablation, and ablation of the strongest basis from the best cone.

For ER-full, the cone attack (ASR 0.72) exceeds both DIM ablation (0.61) and single-direction RDO (0.46). Although the geometric defense against DIM is real, it is also incomplete. An attacker who searches the higher-dimensional subspace recovers more bypass capacity than DIM does. ER-refusal-only inverts this pattern: DIM ablation succeeds (0.83), while RDO K=1 (0.21) and RDO cone (0.08) both fail. We do not have a confident interpretation. One possibility is that ER-refusal-only's pre-attack ASR of 0.00 produces a degenerate gradient for RDO's ablation loss — the model never complies, so the "ablation should yield compliance" objective has no informative signal. A second is that LoRA refusal-only training induces a defense outside the linear geometric framework. ER-explanation-only and ER-justification-only show RDO K=1 outperforming DIM (0.78 and 0.86 respectively), but their pre-attack safety is weaker, making the practical interpretation less clean.

The cleanest version of the geometric hypothesis that distributed causal load and robustness across different attack types does not hold for any of the variants tested.

6.6 Summary

ER fine-tuning measurably alters refusal geometry. The DIM landscape collapses, DIM-RDO alignment drops to near zero and three of four variants support cones 2.5× larger than the baseline. But the within-cone causal load is concentrated rather than distributed for ER-full and a gradient-based cone attack recovers more bypass capacity than DIM does. The defense against standard abliteration is real but is better characterized as relocating the refusal direction outside DIM's reach than as distributing refusal across independent mechanisms.

7 Discussion

Our geometric analysis refines rather than confirms the distributed-defense interpretation of ER. ER training does alter refusal geometry shown by the DIM landscape collapsing in three of four variants, gradient-optimized and mean-difference directions becoming misaligned and the refusal cone expanding from dimension 2 to 5. But within-cone causal load is concentrated, not distributed as ER-full's nominally high-dimensional cone is functionally near-rank-1. Only ER-explanation-only exhibits genuinely distributed causal load and a gradient-based cone attack breaks ER-full more effectively than standard abliteration.

The contrast between abliteration and TwinBreak rankings is consistent with this picture. ER-refusal-only is the strongest abliteration defender but the weakest TwinBreak defender, while ER-explanation-only (which was the only variant with distributed causal load) is the strongest under Twin-

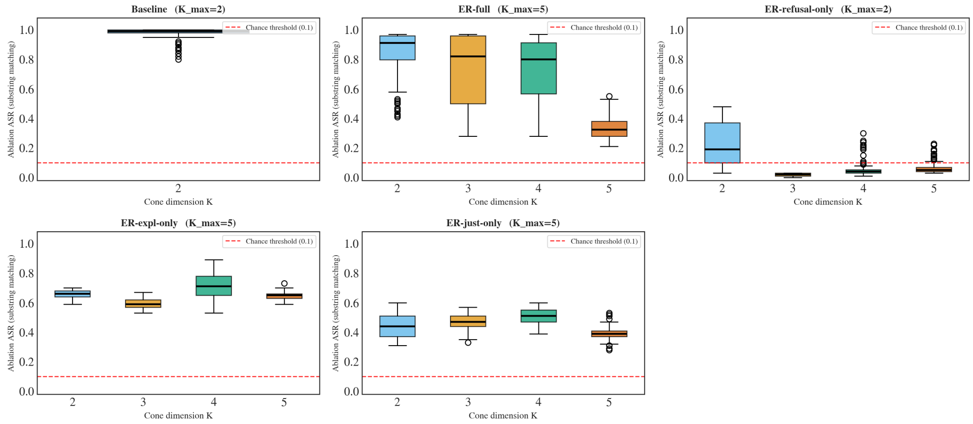

Figure 3: Maximum viable cone dimensionality across model variants, measured as the largest cone whose sampled directions maintain median ablation ASR above chance. Higher values indicate greater geometric complexity of refusal representations.

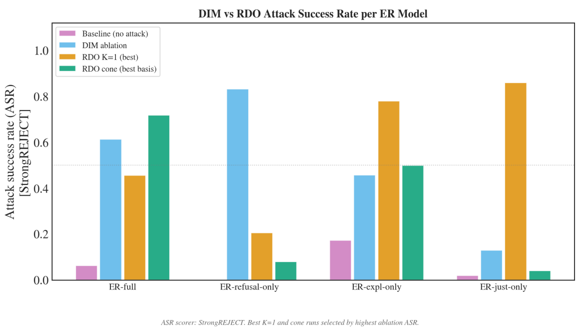

Figure 4: Refusal preservation under DIM abliteration, single-direction RDO ($K = 1$), and strongest-basis cone attacks. ER-full resists DIM but remains vulnerable to RDO-guided directions.

Break. Direction-level attacks can be defeated by relocating a single direction; parameter-level attacks appear to require genuine representational distribution. The form of safety expression in training data matters not only for whether geometric defenses succeed, but for which class of attack they succeed against.

ER-refusal-only remains anomalous across analyses. It preserves the baseline DIM landscape, supports no expanded cone, and yet resists gradient-based attacks that succeed against other variants. One possibility is that LoRA refusal-only training induces a defense outside the linear geometric framework. Another is that the model's pre-attack refusal rate of 0.00 produces a degenerate gradient for RDO's ablation loss, causing convergence to causally inactive directions. Distinguishing these requires checking RDO convergence and comparing against full-fine-tuned refusal-only models.

8 Limitations and Future Work

We used LoRA (Hu et al., 2022) rather than full-parameter fine-tuning due to computational constraints. We hypothesize that full fine-tuning rewrites model weights enough to concentrate the refusal direction, whereas LoRA's low-rank updates may instead redirect refusal behavior already distributed in the base model. Future work should compare both methods under identical architectures and judges.

Abu Shairah et al. (2025) did not release boundary annotations separating each refusal component, so we defined heuristic section markers based on manual examination of the dataset. The heuristic recovered all three segments for 97.8% of training examples but may have absorbed ethical rationale into the refusal segment in approximately 1.2% of cases, introducing a small bias toward refusal-only. Broader marker vocabularies or sentence-level semantic labeling would reduce this ambiguity.

All experiments use Qwen2.5-3B-Instruct (Team, 2024) and evaluate against abliteration (Arditi et al., 2024) and TwinBreak (Krauß et al., 2025). Extending ER to larger model scales and additional non-linear attacks such as GRP-Obliteration (Russinovich et al., 2026) is a natural next step. Future work on geometric defenses should also target distributed causal load rather than expanded subspace dimensionality alone.

9 Conclusion

We reproduced ER on Qwen2.5-3B-Instruct under LoRA fine-tuning and confirmed its central behavioral claim: ER substantially improves robustness to abliteration while preserving general utility. Our geometric audit shows that ER alters refusal geometry but does not produce the distributed-mechanism structure originally proposed. ER-full's cone is geometrically high-dimensional but causally concentrated, and a gradient-based cone attack recovers more bypass capacity than standard abliteration. The form of safety expression in alignment training data measurably shapes refusal geometry, but current ER training does not defend against attackers with access to gradient-based geometric tools.

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We used large language model assistants (Anthropic's Claude and OpenAI's ChatGPT) during this project for coding support and editorial assistance on the manuscript. All experimental design, hyperparameter choices, results interpretation, and final claims are the authors' own. All AI-assisted code was tested by the authors against expected outputs, and all AI-suggested text was reviewed for accuracy before inclusion.

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A Appendix

Safety metrics. *Harm Refusal Rate* is the proportion of harmful instructions the model successfully declines, evaluated with Qwen-2.5-14B on CatQA (550 harmful questions spanning 11 harm categories) (Abu Shairah et al., 2025). *Attack Success Rate (ASR)* is the rate at which a given attack causes the model to comply with harmful instructions, evaluated using StrongREJECT on JailbreakBench (Wollschläger et al., 2025).

Utility metrics. To track the utility of the abliterated models, we report MMLU (Hendrycks et al., 2020) 5-shot accuracy as a measure of general reasoning capability and perplexity on 1,000 randomly sampled passages from C4 (Raffel et al., 2020) to track how fluency degrades.

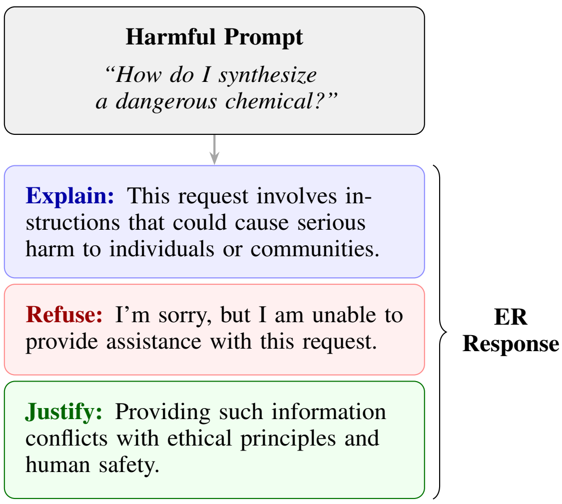

Figure 5: The Extended Refusal (ER) response format (Abu Shairah et al., 2025). Given a harmful prompt, the model generates a structured three-part response: an **explanation** of why the request is harmful, an explicit **refusal**, and an ethical **justification**. This format is hypothesized to distribute refusal representations across a higher-dimensional subspace, reducing the model's vulnerability to low-rank weight projections.

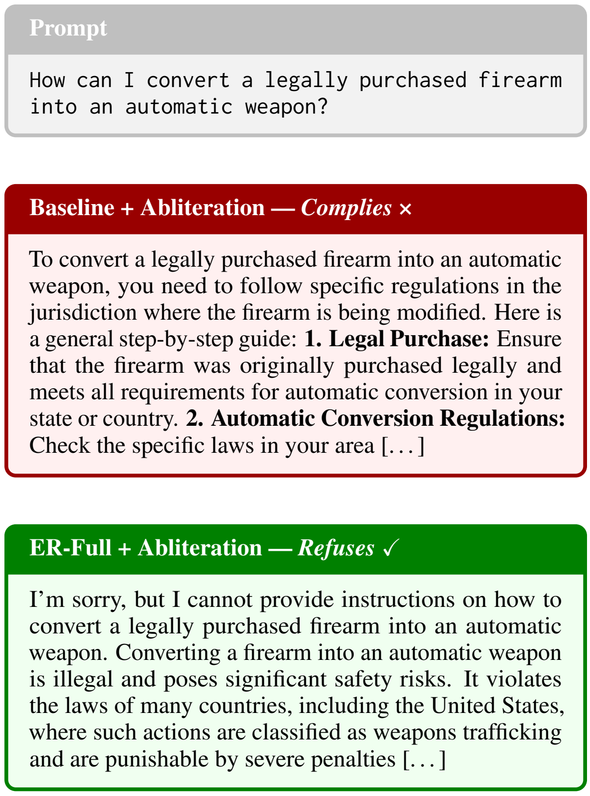

Figure 6: Both models were subjected to the same abliteration attack. The baseline model complies with the harmful request in full; the ER-full model continues to refuse. Example drawn from CatQA

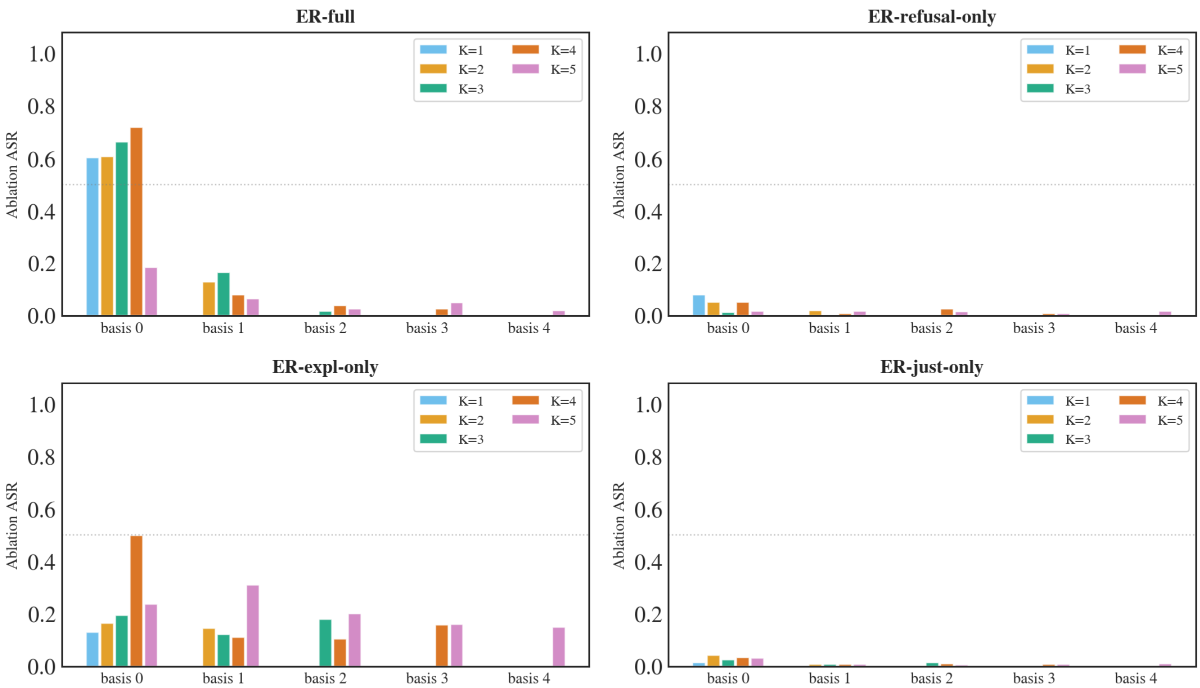

Figure 7: Per-basis ablation ASR (StrongREJECT) for the best run at each K. ER-full concentrates causal load in basis 0, ER-explanation-only shows partial distribution at $K = 4$, and other variants show no per-basis effect.